

FANGPING LAN

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Education

- **Temple University**

Degree: PhD of Computer Science & Information

Major: Computer Science & Information

GPA: 3.7

Anticipated Graduate: June 2026

- **Monmouth University**

Degree: Master of Software Engineering

Major: Software Engineering

GPA: 4.00

Graduate: December 2020

Research Interests

Natural Language Processing (NLP), Large Language Models (LLMs), Agentic AI, Reinforcement Learning (RL), Database Tuning, Prompt Engineer, Machine Learning (ML)

Selected Publications

Google Scholar: <https://scholar.google.com/citations?hl=en&user=K00gXYUAAAAJ>

- **Fangping Lan**, Qi Zhang and Eduard C. Dragut, "Demonstrating CATune: Structural Constraint-Aware Bayesian Optimization for DBMS Configuration Tuning", (VLDB'27 Demo Track, Under Review)
- **Fangping Lan**, Qi Zhang and Eduard C. Dragut, "CATune: Structural Constraint-Aware Bayesian Optimization for DBMS Configuration Tuning", (VLDB'26, Under Review)
- **Fangping Lan**, Qi Zhang and Eduard C. Dragut, "Making Revisions Understandable: A Survey of Edit Intentions, Methods, and Applications", ACL'26 Findings
- Q. Zhang, **Fangping Lan**, C. Caragea, L. J. Latecki, Eduard Dragut "Scaling Performance and Low-Resource Annotation with Many-Shot In-Context Learning for Named Entity Recognition", ACL'26 Findings
- **Fangping Lan**, Abdullah Aljebreen and Eduard C. Dragut, "Why They Link: An Intent Taxonomy for Including Hyperlinks in Social Posts", 18th ACM Web Science Conference 2026 (WebSci'26)
- **Fangping Lan**, Abdullah Aljebreen and Eduard C. Dragut, "One Document, Many Revisions, Too Many Edit Intention Taxonomies", *Findings of the Association for Computational Linguistics (ACL'25)*
- M. Anwar, **Fangping Lan**, A. Wang and M. Caesar, "Indirect Network Troubleshooting with The Chase", June 29-30, (APNet'23)
- **Fangping Lan**, S. Biswas, B. Gui, J. Wu and A. Wang, "Design and Implementation of a Strong Representation System for Network Policies," *2022 International Conference on Computer Communications and Networks (ICCCN 2022)*
- **Fangping Lan**, Bin Gui, and Anduo Wang. 2021. "Fauré: A Partial Approach to Network Analysis". *In Proceedings of the 20th ACM Workshop on Hot Topics in Networks (HotNets '21)*
- Bin Gui, **Fangping Lan**, and Anduo Wang. 2021. "Sarasate: a strong representation system for networking policies". *In Proceedings of the SIGCOMM '21 Poster and Demo Sessions (SIGCOMM '21)*
- Bin Gui, **Fangping Lan**, and Anduo Wang. 2022. "Flexible Routing with Policy Exchange". *In Proceedings of the 5th Asia-Pacific Workshop on Networking (APNet '21)*
- W. Zhang, J. Wang and **F. Lan**, "Dynamic Hand Gesture Recognition Based on Short-Term Sampling Neural Networks", *IEEE/CAA Journal of Automatica Sinica 2020*

Work Experience

- **Applied Scientist II**

Apr. 2026 – now

Company: Amazon.com Services LLC, Amazon JFK27, NY

Project Description: Percolate Optimization Data Platform and Analytics mission is to provide timely and reliable high quality data as well as scalable infrastructure, and tooling to enable the Percolate team to quickly and consistently iterate on innovative approaches to improve customer experience by showing the right content, at the right time, and in the right place for that customer. Our self-service analytics solutions drive better customer experiences and incremental business value by delivering performance insights and

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transparency to all Percolate tenants, including content providers, experience owners, and Percolate internal teams, enabling them to understand content and customer behavior and value on all Percolate powered surfaces.

- **Applied Scientist Intern**

June 2025 – Sept. 2025

Company: Amazon Web Services, Arlington, VA

Project Description: This internship project addresses critical challenges faced by incident responders investigating potential security incidents. Specifically, it targets three key pain points: manual and repetitive search processes across multiple log sources, inefficient search strategies when confronting complex and novel security patterns, and the overwhelming challenge of analyzing large data volumes during time-sensitive investigations. This project will develop an agentic search system that can (1) automate repetitive search tasks across multiple log sources; (2) process natural language queries from security analysts; (3) provide consistent, evidence-based findings.

Skills: LLM, LLM agent, Bedrock API, Strands SDK, Ollama, DuckDB, Python

Project Experience

- **Constraint-Aware Database Tuning**

Sept. 2025 – Mar. 2026

Project description: Developed a database tuning approach that incorporates explicit knob dependency and constraint rules into Bayesian Optimization-based tuning, reducing reliance on pure trial-and-error. By leveraging documented system knowledge, the method avoids invalid or suboptimal configurations and accelerates convergence to high-performance settings.

Skills: Python, GPT-5, Information Retrieval(IR), Bayesian Optimization, Parameters Tuning, PostgreSQL, Benchbase

GitHub: <https://github.com/lanfangping/CATune>

Demo Video: <https://youtu.be/iG0qZ4XV0vQ>

- **Many-Shot In-Context Learning for Named Entity Recognition**

Sept. 2025 – Mar. 2026

Project description: Investigated many-shot in-context learning for Named Entity Recognition, showing that scaling to hundreds of demonstrations enables LLMs to match or surpass fully supervised BERT models. Further demonstrated that many-shot ICL can serve as an effective data annotation framework, generating high-quality labels that yield significant F1 improvements for low-resource NER when used to fine-tune BERT.

Skills: Fine-tune BERT, LLM, NER, Many-Short In-Context Learning(ICL)

- **An Intent Taxonomy for Including Hyperlinks in Social Media Post**

Sept. 2025 – Mar. 2026

Project description: Developed a reader-centered taxonomy of hyperlink intent in social media, combining crowdsourced annotations with LLM refinement. The taxonomy (6 top-level, 26 fine-grained classes) captures diverse communicative purposes and enables intent-aware microblog retrieval, improving content understanding and recommendation.

Skills: Crowdsourcing, LLM, BM25 Retriever, Microblog Retrieval

Website: <https://sites.google.com/view/dmlab-whytheylink>

GitHub: <https://github.com/lanfangping/intent-taxonomy-for-hyperlinked-posts>

- **LLM is A Database Tuning Expert**

Sept. 2024 – Sept. 2025

Project description: Our hypothesis is that LLMs have good capability to tune database since they studied DB tuning expert's experience stored in text, e.g., manuals, blogs, forums, and LLMs can be an agent to interact with DB to optimize configurations. Thus, LLM is a database tuning expert.

Skills: Python, GPT-4, Agentic Reinforcement Learning (RL), Agentic Tuning, PostgreSQL, MySQL, Benchbase

- **One Document, Many Revisions, Too Many Edit Intention Taxonomies**

Sept. 2023 – Jan. 2026

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Project 1 – Edit Intention Taxonomies Construction: We studied existing work to explore the intention behind the edits from the original and revised text. We integrated all proposed edit intention taxonomies into a comprehensive edit intention taxonomy that spans multiple application domains and analytical aspects.

Website: <https://sites.google.com/view/dmlab-unit>

GitHub: <https://github.com/lanfangping/UniT-EditIntentionTaxonomy>

Project 2 – A Survey of Edit Intentions, Methods, and Applications: Conducted the first survey of text revision research from an edit-intention perspective, unifying datasets, taxonomies, identification methods, and applications. Synthesized the end-to-end revision workflow and highlighted key applications and open research challenges in NLP.

- **Pyotr: Tableau as a network representation for deep analysis**

Jan. 2021 – July 2023

Skills: Python, SQL, PostgreSQL, Tableau, Homomorphism

- **Fauré: A partial approach to network analysis**

Sept. 2021 – Dec. 2021

Project description: Fauré, a preliminary design in which a datalog extension (called fauré-log) for incomplete information is developed to enable loss-less modeling, and combined with static analysis of pure datalog to implement example relative-complete verifiers.

Skills: Python, SQL, PostgreSQL, fauré-log

- **Sarasate: A strong representation system for network policies**

June 2021 – Sept. 2021

Project description: we adopt conditional tables and the usual SQL interface (a relational structured developed for incomplete database) as a means to represent and query sets of network states in exactly the same way as a single definite network snapshot.

Skills: Python, SQL, PostgreSQL, conditional tables

Demo: <https://www.youtube.com/watch?v=w9nH2et3zdl> (short version),

<https://drive.google.com/file/d/1KcZKSrbqUhAxfU4tFIE9Vd2WlOt87LW/view> (long version)

Website: <http://ravel-net.org/>

- **Hand Gesture Recognition**

Sept. 2019 – Dec. 2020

Project description: It recognizes the dynamic hand gesture base on a webcam. The hand gesture types include swiping left, swiping right, swiping down, swiping up, pushing hand away, pushing hand in, turning hand clockwise, turn hand counterclockwise, thumb up, thumb down, shaking hand, drumming fingers, stop sign and so on.

Skills: Python, PyTorch, OpenCV, PIL, CUDA, 3D Convolutional Neural Network, Flask, HTML/CSS, JavaScript/jQuery

Project demo video: <https://youtu.be/gBE7cOssUbl>, <https://youtu.be/bSkLH-Ng0D8>

- **Photovoltaic power generation prediction based on LSTM**

Oct. 2018 – May 2019

Project description: Predicting the short-term or ultra-short-term photovoltaic power based on LSTM. This project has been applied into DAQUAN Research Institute.

Skills: Python, HTML, JS, Flask, LSTM, RNN

Training dataset: the dataset of one-year history photovoltaic power and weather condition provided from DAQUAN Research Institute.